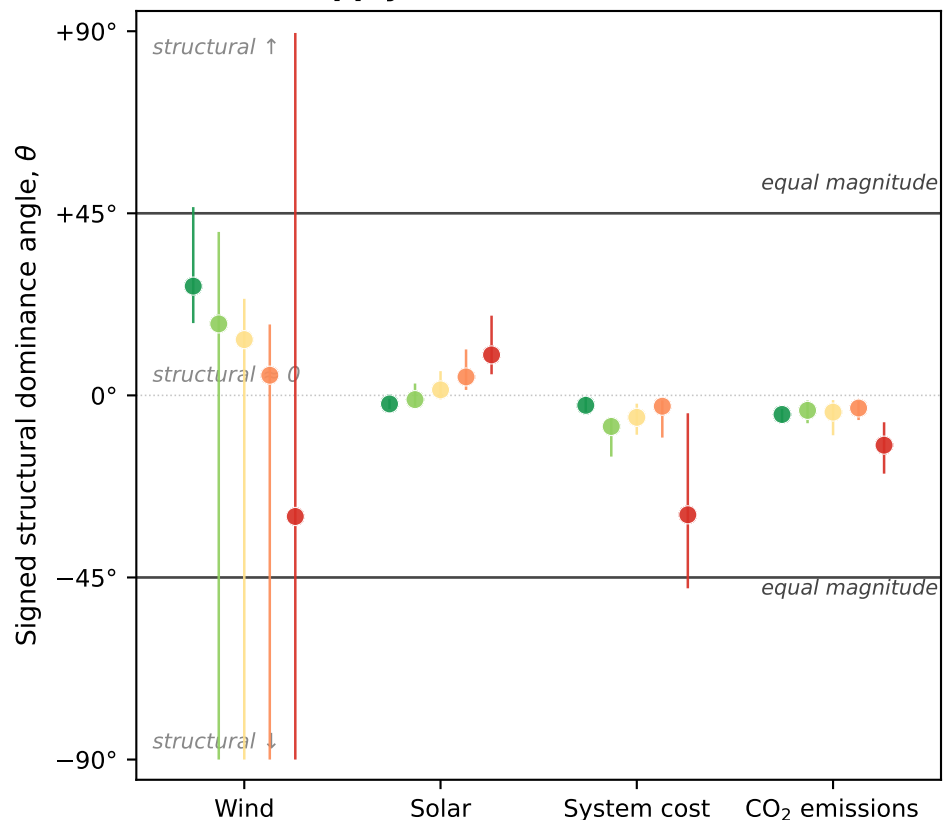


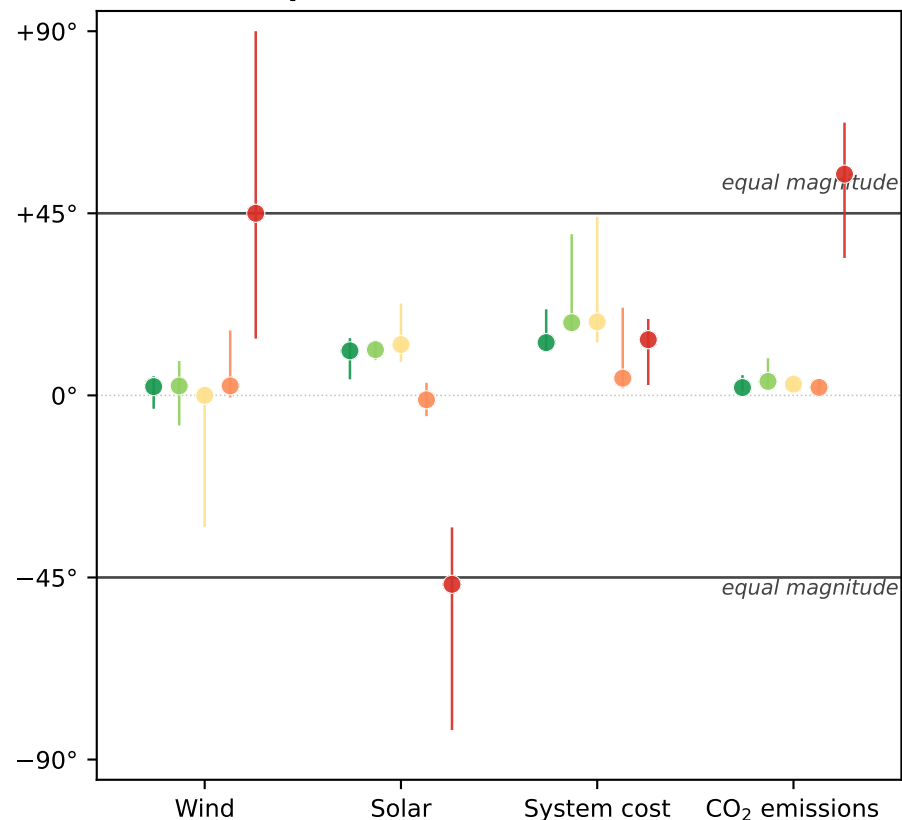
Signed structural dominance across outcomes and climate regimes

Middle East; $\theta = \text{atan2}(\text{structural shift}, |\text{parametric shift}|)$. The sign shows whether the structural shift increases or decreases the outcome; $|\theta| > 45^\circ$ indicates structural shifts larger than parametric shifts. Markers show medians; whiskers show p25 – p75. Anchor scenarios (C7-Base-median, $x = 0$ by construction) are excluded.

Supply-curve detail (LoT→HiT)



Temporal resolution (TS04→TS72)



● C1 (1.5C, low/no OS)

● C3 (likely 2C)

● C7 (above 2.5C)

● C2 (1.5C, high OS)

● C4 (below 2C)

— $|\theta| = 45^\circ$ (structural and parametric shifts equal)

..... $\theta = 0^\circ$ (no structural shift)